

Sunan Zhao

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RESEARCH INTERESTS

- Simulating turbulence using deep learning methods.
- Neural operators integrated with physics-informed methods.
- Large eddy simulation with wall models.

EDUCATION

Southern University of Science and Technology

Ph.D. candidate in Mechanics · Sep 2023 – Present · Shenzhen, China

Supervisor: Prof. Jianchun Wang

China University of Petroleum (East China)

B.Eng. in Engineering Mechanics · Sep 2019 – Jul 2023 · Qingdao, China

PUBLICATIONS

J = Journal, P = Preprint.

[J.2] Zhihong Guo, **Sunan Zhao**, Huiyu Yang, Yunpeng Wang, Jianchun Wang* (2026). *Physics-Informed Transformer operator for the prediction of three-dimensional turbulence*. Acta Mechanica Sinica.

[J.1] **Sunan Zhao**, Zhijie Li, Boyu Fan, Yunpeng Wang, Huiyu Yang, Jianchun Wang* (2025). *LESnets (Large-Eddy Simulation nets): Physics-informed neural operator for large-eddy simulation of turbulence*. Journal of Computational Physics, 537, 114125.

[P.1] **Sunan Zhao**, Yunpeng Wang, Huiyu Yang, Zhihong Guo, Jianchun Wang* (2026). *Large-eddy simulation nets (LESnets) based on physics-informed neural operator for wall-bounded turbulence*. arXiv preprint arXiv:2604.26621. Under review.

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PRESENTATIONS

- LESnets: A physics-informed neural operator for three-dimensional large-eddy simulation of turbulence. Chinese Congress of Theoretical and Applied Mechanics (CCTAM 2025), Changsha, China, 2025. (Oral)
- A physics-informed neural operator for wall-bounded turbulence prediction. The 3rd Chinese Conference of Aerodynamics, Shenyang, China, 2026. (Oral)
- A physics-informed neural operator for wall-bounded turbulence prediction. The 14th National Conference on Fluid Mechanics, Qingdao, China, 2026. (Oral)

HONORS AND AWARDS

China University of Petroleum (East China), College of Pipeline and Civil Engineering, Qingdao

- Second-Class Comprehensive Excellence Scholarship (2020, 2021)
- Outstanding Student Cadre (2020)
- Outstanding Student (2021)

Southern University of Science and Technology, College of Engineering, Shenzhen

- Special-Class Academic Scholarship (2023, 2025)
- Outstanding Graduate Student, University Level (2025)

TEACHING

Undergraduate Teaching Assistant, Southern University of Science and Technology

- MAE 204: Theoretical Mechanics II (Spring 2023)
- OCE 322: Structural Mechanics (Fall 2023)

SKILLS

Programming Languages: Python, Fortran, MATLAB

Language Skills: Mandarin (native), English (fluent)